



CSE-476: Introduction to Natural Language Processing
Quiz 1

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In the BIO tagging scheme for NER, what does the “I-” prefix indicate?
- A.** Beginning of an entity
 - B.** Inside an entity
 - C.** Outside any entity
 - D.** Between two tokens
- Q2. Which statement best describes a (plain) Markov chain in sequence tagging?
- A.** It uses only words (observations) and ignores labels
 - B.** It is identical to an HMM
 - C.** It models transitions between labels but does not consider the words
 - D.** It models emissions $P(\text{word} \mid \text{label})$
- Q3. In a Hidden Markov Model (HMM) for NER, an emission probability is:
- A.** $P(\text{label} \mid \text{previous label})$
 - B.** $P(\text{label} \mid \text{word})$
 - C.** $P(\text{word} \mid \text{previous word})$
 - D.** $P(\text{word} \mid \text{label})$
- Q4. Which of the following best describes Semantic Role Labeling (SRL)?
- A.** A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location
 - B.** A process that assigns grammatical roles like subject and object based on syntactic structure
 - C.** A method that classifies named entities and links them to a knowledge base
 - D.** A technique that labels words with their parts of speech and morphological features

- Q5. Which of the following best describes Byte Pair Encoding (BPE)? **A.** A technique that compresses text by replacing consecutive repeated characters with a count and a symbol
B. A tokenization method that iteratively merges the most frequent pair of symbols in a corpus
C. A tokenization method that splits text based on whitespace and punctuation
D. A character-level encoding scheme that assigns fixed-length binary codes to characters
- Q6. Which of the following best describes the main goal of tokenization in NLP? **A.** To label each token with a grammatical category
B. To convert text into a sequence of symbols suitable for modeling
C. To reduce vocabulary size to zero
D. To assign probabilities to sentences
- Q7. Why is splitting text by spaces (`text.split(" ")`) considered an insufficient tokenizer? **A.** It produces too many tokens
B. It does not work well for languages without explicit word boundaries
C. It cannot tokenize English at all
D. It ignores punctuation entirely
- Q8. Which of the following is NOT listed as a common sequence tagging task? **A.** Part-of-speech tagging
B. Named Entity Recognition
C. Dependency parsing
D. Sentiment analysis
- Q9. What is a common limitation of BIO tagging for Named Entity Recognition? **A.** It cannot represent overlapping or nested entities
B. It requires too many labels
C. It does not support multiple entity types
D. It ignores entity boundaries
- Q10. When choosing between character-level tokenization and sentence-level tokenization, which trade-off must be considered? **A.** Speed vs accuracy
B. Tokenized sequence length vs vocabulary size
C. Precision vs recall
D. Memory vs computation